Exercise 5: 18/06/2026

ADVANCED SQL: NULL Functions

Table 1: Employees

employee_id	name	department	salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

Q1. Show all employees with their salary. If salary is NULL, display 0.
Expected Columns: employee_id, name, salary_with_default.

```
SELECT employee_id,
       name,
       IFNULL(salary, 0) AS salary_with_default
FROM Employees;
```

Output

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2. Show employee names with their department. If department is NULL, show "Not Assigned".
Expected columns: employee_id, name, department_name

```
SELECT employee_id, name,
       IFNULL(department, 'Not Assigned') AS department_name
FROM Employees;
```

output

employee_id	name	department_name
1	Alice	HR.
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2: Orders

order_id	customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3. Find orders with NULL customer_id using ISNULL().
Expected Columns: order_id, customer_id.

```
SELECT order_id,
       customer_id
FROM Orders                OR ISNULL (customer_id);
WHERE customer_id ISNULL;
```

output

order_id	customer_id
103	NULL

Q4. Show all orders. If delivery_date is NULL, show 'Pending'.
Expected Columns: order_id, customer_id, delivery_status.

```
SELECT order_id,
       customer_id,
    IFNULL(delivery_date, 'Pending') AS delivery_status
```

output

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Students

Student_id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5. Show all students and their grades. Replace NULL with 0.
Expected Columns: student_id, name, final_grade.

```
SELECT student_id,
       name,
       IFNULL(grade, 0) AS final_grade
FROM Students;
```

output

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6. Count students who haven't been graded
Expected Columns: not_graded_count

```
SELECT
       COUNT(*) AS not_graded_count
FROM Students
WHERE ISNULL(grade);
```

output

not_grade_count
1

Table 4: Products

product_id	name	price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Q7. Show product name, price, and final price after discount (assume 0 if discount is NULL).
Expected Columns: Product_id, name, final_price

```
SELECT product_id,
       name,
  Price - IFNULL(discount, 0) AS final_price
FROM Products;
```

Output

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503.	Monitor	100

Table 5: Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

Q8. Count how many customers have no email.
Expected Columns: missing_email_count

```
SELECT
      COUNT(*) AS missing_email_count
FROM Customers
WHERE ISNULL(email);
```

Output

missing_email_count
2

Q9. Show all customers with email. If NULL, display "No email".
Expected Columns: customer_id, name, email_display

```
SELECT customer_id,
       name,
    IFNULL( email, 'No email') AS email_display
FROM Customers;
```

Output

Customer_id	name	email_display
1	Linda	No email
2	joseph	joseph@mail.com
3	Nia	No email

Table 6: Payments

payment_id	method	Status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

Q10. Show payment details with method replaced by "Unknown" IFNULL
Expected Columns: payment-id, method_display, status

```
SELECT payment_id,
       Status,
   IFNULL ( method, 'unknown') AS method_dispay
FROM Payments;
```

Output

payment-id	method	Status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

item_id	item_name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11. Show items and their quantity (0 if NULL).
Expected Columns: item_id, item_name, quantity_checked.

```
SELECT item_id,
       item_name,
    IFNULL (quantity, 0) AS quantity_checked
FROM Inventory;
```

Output

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees_Extra

emp_id	bonus	commission
1	NULL	300
2	100	NULL
3	NULL	NULL

Q12. Show employee ID and the first available value among bonus c
commission.
Expected Columns: emp_id, first_available_reward

```
SELECT emp_id,
    ~~IFNOTNULL(~~ COALESCE( bonus, commission) AS first_available
                                                   _reward
FROM Employees_Extra;
```

output

emp_id	
1	300
2	100
3	NULL.

Table 9: Classes

class_id	subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Q13. Count classes that don't have a room assigned.
Expected Columns: no_room_count

```
SELECT
        COUNT(*) AS no_room_count
FROM classes
WHERE ISNULL(room);
```

output

no_room_count
2

Table 10: Attendance

student_id	date	Status
1	2025-04-01	~~[illegible]~~
2	2025-04-01	Present
3	2025-04-01	Absent

Q14. Show attendance records with status. Replace NULL with "Not Marked".
Expected Columns: student_id, date, attendance_status.

```
SELECT student_id, date,
        IFNULL(status, 'Not Marked') AS attendance_status
FROM Attendance;
```

Output

Student id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11 : Bank_Accounts

account_id	account_type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

Q15. Show account ID, account_type (or 'Unknown'), and balance (or 0)
Expected Columns', account_id, type-display, balance_checked

```
SELECT account_id,
 IFNULL (account_type, 'Unknown') AS type_display,
 IFNULL (balance, 0) AS balance_checked
FROM Bank_Accounts;
```

output

account id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 : Projects

project_id	title	start_date	end_date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Q16. Show all projects with a start date. If start_date is NULL, display 'TBD'.
Expected Columns: project_id, title, start_display.

```
SELECT  project_id,
        title,
    IFNULL(start_date, 'TBD') AS start_display
FROM Projects;
```

output

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13: Reviews

review_id	product_id	comment	rating
1	501	Great product	4
2	502	NULL	NULL
3	503	Works fine	3

Q17. Display reviews showing comment (or 'No comment') and rating (or 0)

Expected Columns: review_id, product_id, comment_display, rating_display

```
SELECT review_id,
        product_id,
    IFNULL(comment, 'No comment') AS comment_display,
    IFNULL(rating, 0) AS rating_display
FROM Reviews;
```

Output

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Table 14: Suppliers

supplier_id	name	phone	alt_phone
1	Global Goods	NULL	123456789
2	Best Suppliers	987654321	NULL
3	Vaule Source	NULL	NULL

Q18. Show the supplier contact number. Use COALESCE (phone, alt_phone, 'No Contact').
Expected Columns: supplier_id, name, contact_number

```
SELECT supplier_id,
       name,
    COALESCE (phone, alt_phone, 'No Contact') AS contact_number
FROM Suppliers;
```

Output

Supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Suppliers	987654321
3	Value Source	No contact

Table 15: User_Settings

user_id	theme	Language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC + 1
3	NULL	NULL	NULL

Q19. Show all users and their preferences. Replace all NULLs with defaults:

- Theme → "Light"
- Language → "English"
- Timezone → "UTC"

Expected Columns: user_id, theme_set, language_set, timezone_set

```
SELECT user_id,
       IFNULL(theme, 'Light') AS theme_set,
       IFNULL (language, 'English') AS language_set,
       IFNULL (timezone, 'UTC') AS timezone_set
FROM User_settings;
```

Output

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Table 16: Maintenance

record_id	machine_id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Q20. Show maintenance log with:
- issue → default to "Unknown Issue"
- technician → default to "Not Assigned"

Expected Columns: record_id, machine_id, issue_log, technician_name

```
SELECT record_id,
       machine_id,
   IFNULL(issue, 'Unknown Issue') AS issue_log,
   IFNULL (technician, 'Not Assigned) AS technician_name
FROM Maintenance;
```

Output

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	jammed	Alex